BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 010 · Iteration in C#

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is the difference between = and == in C#?

Q2. Give one reason Unicode may be chosen instead of standard ASCII.

Q3. Convert 01011000 0011 to denary using an 8-bit two’s-complement mantissa and a 4-bit two’s-complement exponent.

Q4. For mark = 85 , what does an if (mark >= 80) ... else if (mark >= 50) structure output, and why?

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I Do · Read new knowledge

Key vocabulary

Recall the term — Iteration

Recall the term — Count-controlled iteration

Recall the term — Condition-controlled iteration

Recall the term — Counter

Recall the term — for loop

Recall the term — while loop

Recall the term — do...while

New learning

Iteration repeats instructions — Iteration means executing a block of instructions repeatedly. The loop needs a rule that determines whether another repetition happens.

for = count-controlled — CODE:
```
for (int i = 0; i < 5; i++)
{
Console.WriteLine(i);
}
```
— The header sets the counter, tests whether the loop continues, then changes the counter after each repetition.

while = test before repeating — A while loop is useful when you do not know how many attempts will be needed. — CODE:
```
while (valid == false)
{
// ask again
}
```

do...while = run first, test after — CODE:
```
do
{
Console.Write("Password: ");
answer = Console.ReadLine();
}
while (answer != "computer");
```
— The body executes before the condition is checked, so it always runs at least once.

Worked example

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Build a times-table program

1. Read the number — CODE:
Console.Write("Number: ");
int number =
int.Parse(Console.ReadLine()); — This happens once before the repetition begins.

2. Count from 1 to 5 — CODE:
for (int i = 1; i <= 5; i++)
{
// repeated instruction
} — i = 1 starts the counter; i <= 5 keeps the loop running; i++ adds 1 after each repetition.

3. Calculate each multiple — CODE:
for (int i = 1; i <= 5; i++)
{
int answer = number * i;
Console.WriteLine(answer);
} — The same two instructions run five times, but i changes each time.

4. Trace a known input — CODE:
number = 7

i = 1 → 7
i = 2 → 14
i = 3 → 21
i = 4 → 28
i = 5 → 35

We Do · Practise with guidance

Output the squares 1^2 to 5^2

Use one for loop. Each repetition should output the square of the current counter.

Complete the pattern

```
for (int i = ?; i <= ?; i++)
{
    Console.WriteLine(?);
}
```

```
for (int i = 1; i <= 5; i++)
{
```

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```
    Console.WriteLine(i * i);
}
```

Validate 1 to 5

Keep asking until the user enters a number from 1 to 5 inclusive.

What controls the loop?

Password with do...while

Ask at least once, then continue until the answer is "computer" .

Reverse the OCR stopping condition

A do...while runs the body before checking the condition.

Describe iteration precisely

An OCR question asks how iteration is used in a function. A strong answer refers to the specific loop , the repeated lines and the condition that keeps it running.

Describe this loop

```
while (number > 0)
{
    number = number / 2;
}
```

```
while (number > 0)
{
    number = number / 2;
}
```

A precise answer names the construct, the repeated code and the continuation condition.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Countdown
Use a for loop to output:
10, 9, 8, ... 1

Question 2

Times-table generator
Ask for a whole number and output its first 10 multiples.

Question 3

Validated age
Keep asking for an age until the user enters a value from 0 to 120 inclusive.
When valid, output Accepted .

Question 4

Password attempts
Use do...while to keep asking until the password is "computer" .
Count how many attempts were made and output the total after the correct password.

Exit Ticket · Answer before checking

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

A while loop always executes at least once.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. = assigns a value. == compares two values for equality.

Q2. Unicode supports a much larger range of characters and writing systems.

Q3. 5.5 Mantissa 01011000 = 0.6875 ; exponent 0011 = 3 ; $0.6875 \times 2^3 = 5.5$.

Q4. Distinction Conditions are tested top to bottom. The first true branch runs, so the later branch is skipped.

You Do pages · Independent questions

Q1. Countdown
CODE:

```
using System;

public class Program
{
public static void Main()
{
for (int i = 10; i >= 1; i--)
{
Console.WriteLine(i);
}
}
}
```

Why it works: the counter starts at 10, continues while it is at least 1, and decreases by 1 after each repetition.

Q2. Times-table generator
CODE:

```
using System;

public class Program
{
public static void Main()
{
```

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

```
Console.Write("Enter a whole number: ");
int number = int.Parse(Console.ReadLine());

for (int i = 1; i <= 10; i++)
{
int answer = number * i;
Console.WriteLine(answer);
}
}
}
```

Why it works: the loop runs exactly ten times and the counter provides the multiplier.

Q3. Validated age

CODE:

```
using System;

public class Program
{
public static void Main()
{
bool valid = false;
int age = 0;

while (valid == false)
{
Console.Write("Enter an age from 0 to 120: ");
age = int.Parse(Console.ReadLine());

if (age >= 0 && age <= 120)
{
valid = true;
}
else
{
Console.WriteLine("Invalid - try again.");
}
}
```

```
Console.WriteLine("Accepted");
}
}
```

Key idea: the Boolean remains false until the accepted condition is met.

Q4. Password attempts

CODE:

```
using System;

public class Program
{
public static void Main()
{
string answer;
int attempts = 0;

do
{
Console.Write("Password: ");
answer = Console.ReadLine();
attempts++;
}
while (answer != "computer");

Console.WriteLine("Attempts: " + attempts);
}
}
```

Key idea: do...while is suitable because the password must be asked at least once. The counter increases on every attempt.

Exit Ticket · Model answer

False. A while loop checks its condition before each iteration and may run zero times.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed: